



CALCULATION PREPARATION

PURPOSE

This document establishes requirements for preparing calculations originated in Fluor offices, Project sites, or other locations including Joint Venture and Workshare offices. Its goal is the preparation of complete and precise calculations with consistency and to reduce overall discipline effort hours.

SCOPE

This practice includes the following major sections:

- APPLICATION
- RESPONSIBILITY
- PREPARATION
- REVISIONS
- ACCEPTANCE CRITERIA
- REFERENCES
- ATTACHMENTS

APPLICATION

This document is to be used by Lead Engineers and Design Engineers involved in the preparation of calculations.

RESPONSIBILITY

Lead Engineer

Consistent with Practice 000.200.0131, Lead Engineers are responsible for:

- Ensuring that calculations are prepared when required, that calculation content is precise and correct and that calculations are recorded and filed for future retrieval (upon job close out) in accordance with the project Records Retention Plan.
- Establishing a method of verification that the project-specific Activity Plan has been followed and implemented for each design item or group of items.
- Develop and issue, when applicable to all staff a listing of readily available structural steel sizes to minimize the number of different shapes used on the project which in turn will help fabricator with mill procurement efforts and minimize warehouse buying costs.



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- Issue to all engineers Form 000.215.F8001 as modified for specific project requirements. Alternatively, incorporate this information within the cover sheet.
- Develop and provide cover pages to be used for each calculation that may include the following information.
 - Client name
 - Project name and location
 - Contract number
 - Calculation number
 - Date and revision number (if required)
 - Originator
 - Number of pages (optional, Lead Structural Engineer to decide)
 - Checking Level
 - Key design assumptions and methodologies
 - Applicable references, codes and standards, with date
 - Professional Engineer's stamp (if applicable)

The Attachments 01 and 02 to this Guideline show samples of Cover Sheet used on recent projects. Project may use this as a guide, with appropriate modifications to suit project needs. Header and footer showing Guideline number shall be deleted when used on the project.

Design Engineer:

Design Engineers are responsible for:

- Knowing, understanding, and implementing the requirements of each assigned task as defined in the Activity Plan, and seeking clarification as required from the appropriate source.
- Understanding all necessary physical interfaces as documented by other disciplines layouts, 3D models, and relevant supplier data.
- Ensuring that the calculation content is precise and correct.
- Provide on every page of each calculation the following information. It may be beneficial to generate a template with the required information to reduce the effort required to provide this for each page.
 - Client name
 - Project name
 - Contract/WBS (work breakdown structure) number, if required



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- Calculation number
- Date
- Originator & Checker (as required for checking level and as assigned by the Lead Engineer)
- Page number
- For calculations with computer analysis, include:
- Hard copy or electronic copy (as directed by project) of input that will enable recreation of model in the future and essential output as defined by Lead Engineer
- Software title and revision used in the calculations
- Graphics, where they might facilitate understanding
- Pertinent output, enough to support member and connection design

PREPARATION

General

Prepare calculations in a professional, well-organized, and cost-effective manner. See Practice 000.200.1400. When preparing calculations ensure familiarity with the following:

- Baseline documents, particularly the execution strategy, estimate (man-hours and materials quantity) and schedule
- Activity Plan, Work Instructions, Checklists attached to 000.200.1020, and Criticality Rating System
- Design Criteria
- Construction Specifications
- Standard Drawings
- Applicable building codes, reference systems and design guidelines
- Project specific requirements (for example limited steel shapes, rebar sizes, etc)

The design basis is to be consistent with the initial layout which is typically established in consultation with other disciplines. Check with Civil, Architectural, Mechanical, Piping and Electrical for any restricting features such as bracing locations, clearance beneath truss or beam, and depth to top of foundations versus underground utilities. The final calculations must adequately validate the final drawings.

Fluor guideline sample designs typically show every step in a problem solution and are intended to provide detailed guidance to engineers. It may not be necessary to prepare calculations to the same level of detail as the guidelines. Experienced engineers are expected to develop a feel for what will be critical and calculate only critical cases, bypassing others by addition of words like "Okay by Engineering Judgment."



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Unless required for submittal to the client or the authority having jurisdiction, discuss need more documenting formal calculations for routine lightly loaded pipe supports, electrical supports, cross-overs and stiles, and other minor items which can be sized by experience.

Avoid excessive refinement of computations. Precision of calculations should be consistent with accuracy of input data. Reasonable approximations and short-cuts in computations are justifiable to save effort hours and schedule time.

Unique details and calculations should be submitted to the CSA Knowledge Community for the future reference.

Calculations shall mention all assumptions of the design and demonstrate that the intended design is fit for purpose.

Calculations shall be will sequenced and coordinated. Reference shall be provided at appropriate locations. Third party documents (Vendor drawings, calculations...etc.) where required shall be included partly or as a whole in calculations for easy referencing.

Computer-Generated Calculations

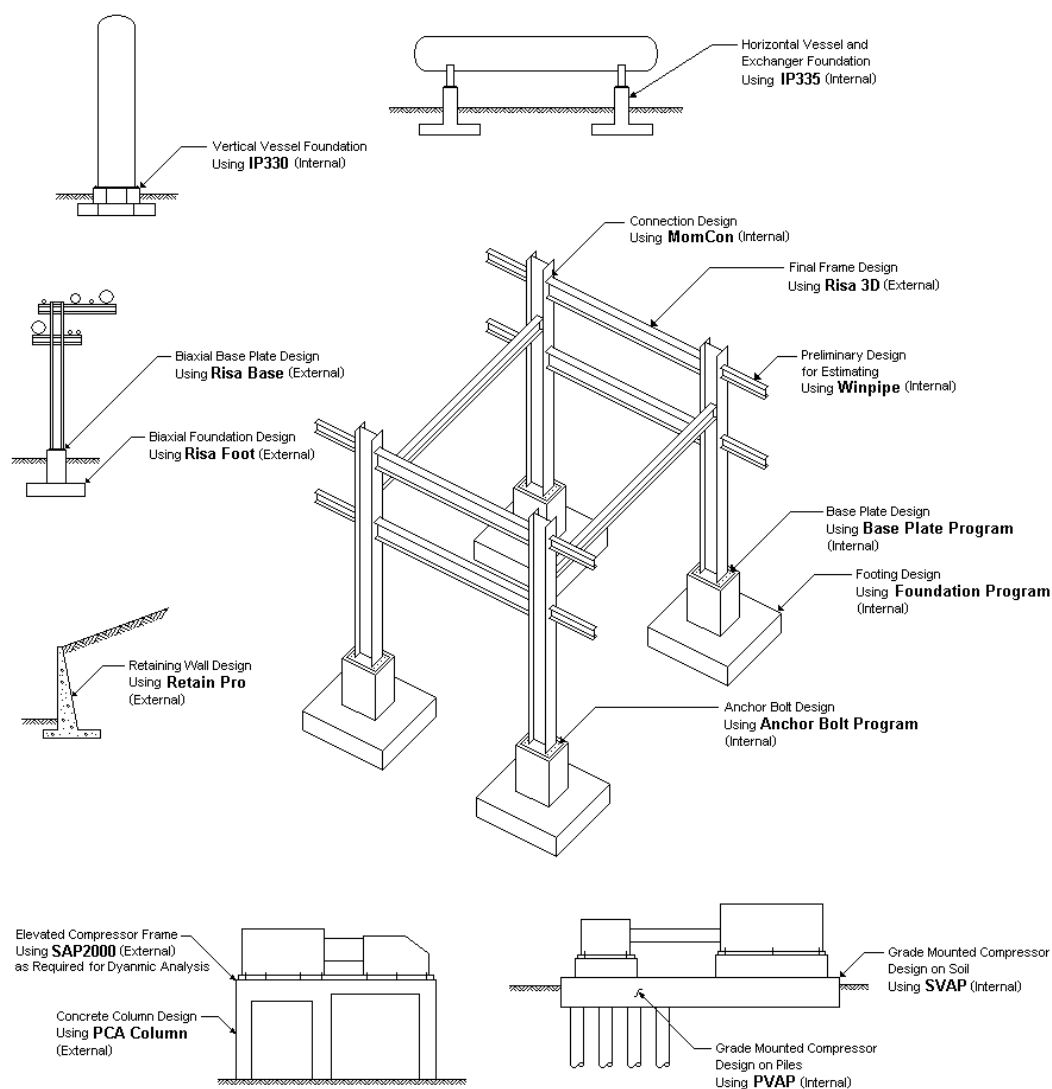
Calculations may be prepared using spreadsheets, third-party software, or Fluor reference system software. Use computer generated calculations to increase calculation accuracy and efficiency, not to make them "pretty". Do not use computer software when a manual design will be quicker, unless there is reason to believe that the calculations will be repeated for similar items.

When using analysis/design software for large complex structures:

- Modeling of every interior member and loading of every small piece of equipment is not necessary. Uniform loadings on main framing members may be sufficient to provide suitable design loads for the main beams, columns, lateral force resisting system, and foundations. Design of fill beams and equipment supports can be performed by manual calculations. As a general guideline, model the systems that make up the primary vertical and lateral load paths. Both the mass and stiffness must be precisely modeled for dynamic analyses.
- Do not run the program "one last time" to include small changes. It is acceptable to provide manual annotations to note the drawing variance from the computer output.
- Limit the number of load combinations to what is necessary.



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See the figure above for a useful way of summarizing the various third-party and Fluor reference system global software, and how they can be applied to the design of typical structures and foundations on a project.

Engineering Sketches

Provide preliminary sketches, as required, to the Design Supervisor noting overall configuration and member sizing for use in developing the 2D drawings and 3D model.



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Throughout the calculation process, keep the Design Supervisor informed of any changes. Although potential cycling of updates and changes to the 3D model during design development may appear inefficient to the design effort, an up-to-date 3D model is absolutely necessary for the other disciplines and allows timely interference checking, resulting in an overall savings to the project.

Prepare neat engineering sketches to clearly provide the amount of information needed to complete the design drawing, including relevant Supplier requirements. Engineering sketches are not an end product and they don't need to have the quality and look of the finished drawing. Spending time on unnecessary enhancement of sketches is a duplication of effort and sacrifices efficiency. It is not necessary to revise the engineering sketch just to match the drawing. Use of "fill-in-the-blank" sketches is encouraged to save effort-hours and schedule time. Complex steel connection or concrete reinforcing sketches may be required to be drawn to scale to validate practicality. Mark-ups of preliminary drawings can also be used to convey changes and additions.

Maximize the use of standard details. If a design condition does not exactly match a standard detail, the Engineer should consider reference to a standard detail with exceptions noted (for example, "Detail No. 27, similar, with 3 bolts"). Engineering sketches of details should be minimized.

REVISIONS

Revisions to completed calculations will be made when the Lead Engineer considers that the assumptions utilized in the original calculations are no longer valid or that the original conditions have changed. The Originator will make revisions, where possible. All revisions made to a set of completed calculations shall be clouded, initialed, and dated by the engineer making the modifications. Update the cover sheet to reflect the revision and state the reason for change.

ACCEPTANCE CRITERIA

The signature of the originator on the calculation cover sheet, calculations, and design sketches indicate compliance with this document.

REFERENCES

General Engineering

[Practice 000.200.0131](#) Lead Engineers – Roles and Responsibilities

[Practice 000.200.1400](#) Cost Competitive Engineering and Design



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RESOURCES

General Engineering

Practice 000.200.0077	Criticality Rating System
Practice 000.200.1020	Calculation Checking
Practice 000.200.1200	Designing to the Capital Cost Estimate
Practice 000.200.1401	Design Margin Table

Civil/Structural/Architectural Engineering

Practice 000.215.F0072	Activity Plan: Civil/Structural Engineering
Guideline 000.215.0760	Fireproofing Requirements for Structural Steel Members
Guideline 000.215.0965	Guidelines for Field Investigations for Revamp Projects
Guideline 000.215.1201	Single Plate Connections
Guideline 000.215.1203	Bracing Connections
Guideline 000.215.1208	Base Plate Design Criteria
Guideline 000.215.1209	Bolted End Plate Moment Connections
Guideline 000.215.1215	Wind Load Calculation
Guideline 000.215.1216	Earthquake Engineering
Guideline 000.215.1910	Work Instructions: Calculations
Guideline 000.215.F8001	Structural Engineering Calculations Computer File Index
Guideline 000.215.STE05121	Anchor Bolt Design Criteria
Specification 000.215.STC01015	Structural Design Guide
Specification 000.215.STI03310	Concrete Typical Details
Drawing 000.215.4060	Concrete Fireproofing of Structural Steel – Typical Details



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<u>Drawing 000.215.4061</u>	Pyrocrete Fireproofing of Structural Steel – Typical Details
<u>Drawing 000.215.5030</u>	Structural Steel General Notes
<u>Drawing 000.215.5040</u>	Structural Steel Connection Details (S1 thru S24)
<u>Drawing 000.215.5050</u>	Structural Steel Connection Details (S30 thru S45)
<u>Drawing 000.215.5060</u>	Structural Steel Connection Details (S52, Erection and Bracing Connections)
<u>Drawing 000.215.5230</u>	Steel to Concrete - Typical Details
<u>Reference 000.215.9885</u>	Steel Connections - Details and Relative Costs
ACI 318/ACI318M	American Concrete Institute - Building Code Requirements for Structural Concrete
ANSI/AISC 360	American Institute of Steel Construction - Specification for Structural Steel Buildings
	Other Applicable Building Codes and Standards

ATTACHMENTS

Attachment 01	Cover Sheet Sample 01
Attachment 02	Cover Sheet Sample 02



CALCULATION PREPARATION

(SAMPLE 01)



MARATHON OIL COMPANY

DETROIT HEAVY OIL UPGRADE PROJECT

CONTRACT No.: A2IS

DESIGN OFFICE: IRVINE

DISCIPLINE: STRUCTURAL

DESIGN CALCULATION No: A2IS- AABBC-DD-EE
(See EC-00-01 Sect 2.1B)

TITLE: _____

CHECKING LEVEL: _____

(Engineer to verify appropriate checking level w/ Practice 000.215.1910, Attachment 01)

REV:	NO. OF PAGES	BY/DATE:	CHK'D/DATE:
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_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____



CALCULATION PREPARATION

MARATHON OIL COMPANY
DETROIT HEAVY OIL UPGRADE PROJECT
Detroit, MI
CONTRACT NO. A2IS



By: _____ Date: _____
Chkd: _____ Date: _____
Calc No.: A2IS-AABBC-DDEE

CALCULATION TABLE OF CONTENTS

<u>Item:</u>	<u>Page No.:</u>
Calculation Cover Page	
Engineering Sketches.....	SK1-SKXX
Calculation table of Contents	i
Calculation Computer File Index	ii
Introduction and Lateral Force Resisting System Description	iii
Basic Design Data.....	iv
Calculations.....	1
XXX.....	X
Attachment.....	Attach A, Etc.



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MARATHON OIL COMPANY
DETROIT HEAVY OIL UPGRADE PROJECT
Detroit, MI
CONTRACT NO. A2IS



By: _____ Date: _____
Chkd: _____ Date: _____
Calc No.: A2IS-AABBC-DDEE

CALCULATION COMPUTER FILE INDEX

Calculation No: A2IS-AABBC-DD-EE

The purpose of this form is to keep track of calculation computer files and to assign filenames using a standard format. This will assist in organizing data at a later date. The file naming scheme shown below should be used unless specific project requirements dictate otherwise

Include this form with each set of calculations which has computer files. List all data files that were used to complete the calculations. Place all electronic files in the appropriate sub directory under A2IS03/215/calculation. Filenames shall be as follows:

AABBC-DD-RR.ZZZ

Where:

AA – Unit Number
BB – Discipline Identifier
C – Designates Calculation
DD – Calc Sequence Number
RR – Run Number
.ZZZ – .Extension given by application



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MARATHON OIL COMPANY
DETROIT HEAVY OIL UPGRADE PROJECT
Detroit, MI
CONTRACT NO. A2IS

FLUOR®
By: _____ Date: _____
Chkd: _____ Date: _____
Calc No.: A2IS-AABBC-DDEE

File Name	Description	Directory Location



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MARATHON OIL COMPANY
DETROIT HEAVY OIL UPGRADE PROJECT
Detroit, MI
CONTRACT NO. A2IS

FLUOR[®]
By: _____ Date: _____
Chkd: _____ Date: _____
Calc No.: A2IS-AABBC-DDEE

INTRODUCTION AND LATERAL FORCE RESISTING SYSTEM **DESCRIPTION:**

The introduction page should include the following items:

1. A brief overall description of the structure/equipment you are designing/supporting.
2. Overall dimensions, height number of levels, number of bays, etc.
3. A description of the lateral force resisting system in each direction. Type of braced or moment frame system such ordinary or special and assumed seismic R factors.
4. Basic assumptions such as fixed or pinned bases.

Please remember that the building Department engineers who may be reviewing your calculations for the first time and do not have the benefit of plot plans, smart plan viewer, or any other drawings of the project. The better description you provide the easier it will be for the engineer to review your work.



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DETROIT HEAVY OIL UPGRADE PROJECT
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FLUOR
By: _____ Date: _____
Chkd: _____ Date: _____
Calc No.: A2IS-AABBC-DDEE

BASIC DESIGN DATA

(BASED ON MASTER SPECIFICATION EC-00-01)

DESIGN CODES:

2003 MICHIGAN BUILDING CODE (MBC)
ASCE 7-05 MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES
ACI 318-05 BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE
AISC 360-05 SPECIFICATION FOR STEEL BLDGS.
AWS D1.1-04 STRUCTURAL WELDING CODE
SPECIFICATION FOR STRUCTURAL JOINTS USING ASTM A325 OR ASTM A490 BOLTS.

SNOW LOADING

MBC/ IBC SECTION 1608 AND ASCE 7-05, SECTION 7
GROUND SNOW LOAD = 30 psf
I = 1.1 (ASCE 7-05, TABLE 7-4 for Occupancy Category III)

WIND LOADING

MBC/ IBC SECTION 1609 AND ASCE 7-05, SECTION 6
BASIC WIND SPEED = 90 mph (3 Sec Gust)
EXPOSURE = C
 $I_w = 1.15$ (ASCE 7-05, TABLE 6-1 for Occupancy Category III)

EARTHQUAKE LOADING

MBC/ IBC SECTION 1613 AND ASCE 7-05 (Except for Chapter 14 and appendix 11A)
 $S_{DS} = 0.21g$ $S_s = 0.126g$
 $S_{D1} = 0.11g$ $S_1 = 0.046g$
SEISMIC SITE CLASS = CLASS E
SEISMIC DESIGN CATEGORY B
I = 1.25 (SEISMIC IMPORTANCE FACTOR ASCE 7-05 TABLE 11.5-1 for Occupancy Category III)



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$I_p = 1.5$ (SEISMIC IMPORTANCE FACTOR for nonstructural components containing hazardous materials or required for life safety)

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DETROIT HEAVY OIL UPGRADE PROJECT
Detroit, MI
CONTRACT NO. A2IS

FLUOR
By: _____ Date: _____
Chkd: _____ Date: _____
Calc No.: A2IS-AABBC-DDEE

MATERIAL PROPERTIES

CONCRETE:

$f'_c = 4000\text{psi}$

REINFORCING STEEL

ASTM A615 GRADE 60

STRUCTURAL STEEL – UNLESS SPECIFICALLY NOTED OTHERWISE

WIDE FLANGE SHAPES AND PLATES

ASTM A992 GRADE 50

CHANNELS & ANGLES

ASTM A36

BOLTS

HIGH STRENGTH BOLTS – ASTM A325

ANCHOR BOLTS – UNLESS SPECIFICALLY NOTED OTHERWISE

ASTM F1554 GRADE 36

ALLOWABLE NET SOIL BEARING

REFER TO GEOTECHNICAL REPORT

ALLOWABLE PILE CAPACITY

REFER TO DWG SD-929

OTHER REFERENCE DATA



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(Option to Insert Project Logo here rather than or in addition
to having Client and Location/ Project Name below)

SAMPLE 02

Civil Structural Calculation Cover Pages

Client and Location: <<Insert>>
Project Name: <<Insert>>
Project Number: <<Insert>>
Calculation Number: <<Insert>>

<<A project specific template of this form is to be saved in the following project folder for use by the team.
\\215_CSA\215E Engineering\215E.14R Calculations>>

Calculation Title: <<Insert>>

Checking Level: <<Insert>>

Drawing Numbers - Titles:

Steel - <<Insert first steel drawing of series>>

Concrete - <<Insert first concrete drawing of series>>

Design Specifications:

Civil Design Criteria: See specification ...

Structural Design Criteria: See specification ...

Approvals for First "Issue for Construction" of the Design

	Name (Print)	Signature	Date (yr/mo/day)
Originator			
Checker			
Engineer of Record			

Record of Revisions

Rev. No.	Reason for Revision (Only for changes after first issue)	Date (yr/mo/day)	Engineer	Checked By



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(Option to Insert Project Logo here rather than have the Client and Location/ project name below)

Civil Structural Calculation Cover Pages

Contract Number: <<Insert>>
Calculation Number: <<Insert>>
Date: <<ddMmmyy>>

Software Name/version:

<<Insert>>

<<Insert additional as required>>

(Ensure software is on the project software matrix. Could be accomplished by listing the software matrix contents on the template and have the users delete unused software)

Electronic File Names:

File Name	Description	Directory Location

References used:

<<Insert>>

<<Insert additional as required>>

Strength and Serviceability Design Requirements:

<<Types of loads, deflection limits, special considerations>>

<<The majority of this information should be a standard on a given project>>

Design Load Basis: (Non-typical items or items requiring selection from options in the design specifications only)

Platform Live Load: <<Describe each specific item>>

Others: (If applicable)

Deflection Limits:

<<Describe each deflection limit including references to applicable bulletins>>

Vendor Data:

<< State one of the following: (repeat as required for each piece of equipment):

- Final load information was available and is the basis of design.
- No equipment loads will act on this structure.



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- If neither of the above apply then describe the equipment load basis for design and refer to the Design Requirements Uncertainties List>>

Equipment Tag Number	Document Number	Revision

Pipe Stress Information:

<<State one of the following:

- Final Pipe Stress information was available and is the basis of design.
- No Pipe Stress loads will act on this structure.
- If neither of the above apply then describe the Pipe Stress basis for design and refer to the Design Requirements Uncertainties List.>>

Other Design Requirements: (Describe each criteria – Examples of potential items listed below)

<< Vibration Limits >>

<< Loading on partially erected equipment structures >>

<< Module Transportation and Lifting >>

<< Others >>



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Design Requirement Uncertainties (Risk) and Mitigation:

<<Items will also be listed in an Overall Discipline Uncertainty and Risk log>>

Refer to JDB 6.99 for more information and an extensive list of example Uncertainties and Mitigations.

Uncertainty	Mitigation (Design Requirement)	Remaining Risk	Comments	Resolution of Uncertainty



CALCULATION PREPARATION

Table of Contents (Only required for larger calculations. Example provided for a Steel Structure can be altered and expanded)

Section	Page Number
Structure/Foundation Description, Calculation Purpose, Sketches	1.x
Analysis Model Considerations	2.x
Load Development, Application, and Combinations	3.x
Analysis Results (Reactions and deflections)	4.x
Connection Design	5.x
Member Design	6.x
Foundation Design	7.x

Appendix (reference documents and other supporting documents)

Vendor Documents
Pipe Stress Information
3D Model Screen Prints